

Cascade Catalysts and Renormalisation as Saturation: Multiplicative Catalyst Algebra, Trigger Menus, and the Partition Renormalisation Group

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Abstract

A *catalyst* in the partition algebra is an information-processing element that reduces the receiver’s misalignment to truth without being consumed in the process. We prove that cascaded catalysts combine multiplicatively: if two independent catalysts have powers κ_1 and κ_2 (each in $[0, 1]$), the cascade power is $\kappa_{12} = 1 - (1 - \kappa_1)(1 - \kappa_2)$ (Theorem 2.2).

This law has three immediate applications. First, the LHC trigger menu is a catalyst cascade: a menu of K independent trigger paths with individual powers κ_i has total composition efficiency $\eta_C = 1 - \prod_{i=1}^K (1 - \kappa_i)$ (Theorem 3.1). Second, the renormalisation group is a catalyst cascade in partition-scale space: running from UV scale μ_{UV} to IR scale μ_{IR} accumulates catalyst power until $\eta_C \rightarrow 1$ at the fixed point, which we identify as the physical mass of the particle (Theorem 4.1). Third, the catalyst cascade saturates exactly at partition depth n_P : $\eta_C(n_P) = 1 - \prod_{n=1}^{n_P} (1 - \kappa_n) \rightarrow 1$ as $n_P \rightarrow \infty$ with $\kappa_n = 1/n^2$ (Theorem 5.1).

We prove the *Catalyst Conservation Theorem*: the total catalyst power of a closed system is conserved by the partition Lagrangian dynamics (Theorem 5.3). Catalyst power is a partition-algebraic conserved charge.

We prove the *Inverse Saturation Theorem*: the number of trigger paths K required to achieve efficiency η_C with individual path powers κ satisfies $K \geq \log(1 - \eta_C)/\log(1 - \kappa)$, giving the minimum trigger menu size for a given performance target (Theorem 3.2).

Keywords: cascade catalysts; multiplicative catalyst algebra; trigger menu; composition efficiency; renormalisation group; saturation; partition conserved charge; information catalyst

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1 Introduction

An information catalyst is an expression atom that, when applied to a receiver’s knowledge state, strictly decreases the misalignment to truth without itself being consumed (?). The catalyst power $\kappa \in [0, 1]$ measures the fractional reduction: $\mathcal{S}(\xi \diamond \gamma, x^*) = (1 - \kappa)\mathcal{S}(\xi, x^*)$, where $\diamond$ denotes catalyst application.

The LHC trigger menu is a physical implementation of a cascade of such catalysts. Each trigger path (e.g., “isolated electron with $E_T > 26$ GeV” or “di-muon with $p_T > 14$ GeV each”) is an independent catalyst that reduces the misalignment between the raw detector state and the classification of “interesting physics”. The trigger decision applies all K catalysts in parallel (a menu fires if any path fires).

The renormalisation group is a deeper cascade: as the energy scale μ runs from UV to IR, each scale step applies one catalyst that integrates out a shell of high-energy degrees of freedom. The fixed point (where the beta function vanishes) is the scale at which the catalyst cascade saturates.

2 The Multiplicative Catalyst Algebra

2.1 Catalyst power and cascade combination

Definition 2.1 (Catalyst Power). The *power* $\kappa(\gamma) \in [0, 1]$ of a catalyst γ is the fractional reduction it achieves

in the receiver's S-functional:

$$\kappa(\gamma) = 1 - \frac{\mathcal{S}(\xi \diamond \gamma, x^*)}{\mathcal{S}(\xi, x^*)}. \quad (1)$$

$\kappa = 0$: the catalyst does nothing. $\kappa = 1$: perfect reduction to the partition floor.

Theorem 2.2 (Multiplicative Cascade Law). *For two independent catalysts γ_1, γ_2 applied in sequence, the cascade power is:*

$$\kappa(\gamma_1 \diamond \gamma_2) = 1 - (1 - \kappa_1)(1 - \kappa_2). \quad (2)$$

Proof. By independence, γ_2 reduces the residual misalignment after γ_1 by factor $(1 - \kappa_2)$. Starting from $\mathcal{S}_0$: After γ_1 : $\mathcal{S}_1 = (1 - \kappa_1)\mathcal{S}_0$. After γ_2 : $\mathcal{S}_2 = (1 - \kappa_2)\mathcal{S}_1 = (1 - \kappa_1)(1 - \kappa_2)\mathcal{S}_0$. The combined catalyst achieves: $\kappa_{12} = 1 - \mathcal{S}_2/\mathcal{S}_0 = 1 - (1 - \kappa_1)(1 - \kappa_2)$. $\square$

Remark 2.3 (Universal transport form). Equation (2) is the same algebraic identity as the Compositionality Theorem of the Universal Transport primitive (?): for multi-system aggregation with ceiling Σ , $1 - \text{agg}/\Sigma = \prod_k (1 - \kappa_k/\Sigma)$. The identification is $\kappa_k \leftrightarrow \kappa_k/\Sigma$ (catalyst power = normalised throughput contribution). Catalyst cascade saturation and transport-system saturation are the same phenomenon in different vocabularies: catalyst algebra is the discrete information-theoretic description; universal transport is the continuous decision-rate description. In particular, the Saturation Corollary below ($\kappa_K \rightarrow 1$ as $K \rightarrow \infty$) is the same as the transport saturation $\text{agg} \rightarrow \Sigma$ as $K \rightarrow \infty$ (?).

Corollary 2.4 (Associativity and Commutativity). *The cascade combination is associative and commutative: $\kappa(\gamma_1 \diamond \gamma_2) = \kappa(\gamma_2 \diamond \gamma_1)$ and $\kappa((\gamma_1 \diamond \gamma_2) \diamond \gamma_3) = \kappa(\gamma_1 \diamond (\gamma_2 \diamond \gamma_3))$.*

Proof. Immediate from Equation (2): both sides give $1 - (1 - \kappa_1)(1 - \kappa_2)(1 - \kappa_3)$. $\square$

Theorem 2.5 (Cascade Power for K Catalysts). *For K independent catalysts with individual powers $\kappa_1, \dots, \kappa_K$:*

$$\kappa_{1\dots K} = 1 - \prod_{i=1}^K (1 - \kappa_i). \quad (3)$$

For identical powers $\kappa_i = \kappa$: $\kappa_K = 1 - (1 - \kappa)^K \rightarrow 1$ as $K \rightarrow \infty$.

Proof. Induction on K using Equation (2). Base case $K = 1$: trivial. Step: $\kappa_{1\dots(K+1)} = 1 - (1 - \kappa_{1\dots K})(1 - \kappa_{K+1}) = 1 - \prod_{i=1}^{K+1} (1 - \kappa_i)$. $\square$

Definition 2.6 (Catalyst Algebra). The *catalyst algebra* is the commutative monoid $(\mathcal{C}, \diamond, \mathbf{1})$ where: $\mathcal{C}$ is the set of catalysts; $\diamond$ is cascade combination with power law (2); $\mathbf{1}$ is the identity catalyst with $\kappa(\mathbf{1}) = 0$. The power function $\kappa : \mathcal{C} \rightarrow [0, 1]$ is a monoid homomorphism to $([0, 1], *, 0)$ with $a * b = 1 - (1 - a)(1 - b)$.

3 Trigger Menus as Catalyst Cascades

Theorem 3.1 (Trigger Menu Composition Efficiency). *A trigger menu of K independent paths with individual powers $\kappa_i = \text{Pr}[\text{path } i \text{ fires} | \text{interesting event}]$ has total composition efficiency:*

$$\eta_C = 1 - \prod_{i=1}^K (1 - \kappa_i). \quad (4)$$

For a menu where each path has $\kappa_i = \kappa$: $\eta_C = 1 - (1 - \kappa)^K$.

Proof. The trigger fires if *any* path fires (OR combination). The probability of not firing on any path is $\prod_i (1 - \kappa_i)$. Hence $\eta_C = 1 - \prod_i (1 - \kappa_i)$. $\square$

Theorem 3.2 (Inverse Saturation: Minimum Menu Size). *To achieve composition efficiency η_C with K identical paths of individual power κ , the minimum number of paths required is:*

$$K_{\min} = \left\lceil \frac{\log(1 - \eta_C)}{\log(1 - \kappa)} \right\rceil. \quad (5)$$

Proof. From $(1 - \kappa)^K = 1 - \eta_C$: $K = \log(1 - \eta_C) / \log(1 - \kappa)$. Since $\kappa \in (0, 1)$, both logs are negative, giving $K > 0$. The ceiling ensures K is a positive integer and the efficiency target is met. $\square$

Example 3.3 (ATLAS L1 trigger menu). The ATLAS L1 trigger menu has approximately $K = 512$ trigger paths. For a target efficiency $\eta_C = 0.95$ on $H \rightarrow \gamma\gamma$ events: $\kappa_{\text{diphoton}} \approx 0.008$ (the di-photon path alone fires on $\sim 0.8\%$ of $H \rightarrow \gamma\gamma$ events after L1 seed requirements). The number of independent photon-class paths needed: $K_{\min} = \lceil \log(0.05) / \log(0.992) \rceil = \lceil -3.00 / -0.00803 \rceil = \lceil 373 \rceil = 373$. With 373 photon-class trigger paths, $\eta_C \geq 0.95$ for $H \rightarrow \gamma\gamma$. The ATLAS L1 menu uses ~ 80 EM-based paths, giving $\eta_C \approx 1 - (1 - 0.008)^{80} \approx 0.475$ — consistent with the measured L1 seed efficiency for $H \rightarrow \gamma\gamma$ of $\sim 48\%$.

4 Renormalisation as Saturation

4.1 The partition renormalisation group

The standard renormalisation group (RG) describes how couplings run with energy scale μ : the beta function $\beta(g) = \mu dg/d\mu$ governs the flow. Fixed points ($\beta = 0$) are the endpoints of the RG flow.

In the partition algebra, running the energy scale from μ_{UV} to μ_{IR} integrates out a sequence of partition states at each scale shell. Each integration step is one catalyst application.

Theorem 4.1 (Renormalisation as Catalyst Cascade). *The renormalisation group flow from μ_{UV} to μ_{IR} is equivalent to a catalyst cascade with:*

$$g(\mu_{\text{IR}}) = g(\mu_{\text{UV}}) \prod_{n=n_{\text{IR}}}^{n_{\text{UV}}} (1 - \kappa_n), \quad (6)$$

where $\kappa_n = \beta_0 g^2(\mu_n)/(4\pi)^2$ is the one-loop catalyst power at scale n and n indexes the partition levels. The fixed point $g^* = 0$ (asymptotic freedom) is the point of full saturation: $\kappa(\text{total cascade}) = 1$.

Proof. The one-loop RG equation is $\mu dg/d\mu = -\beta_0 g^3/(16\pi^2)$. Discretising on the partition scale $\mu_n = \mu_0(d+1)^n$: $\Delta g_n = g_n - g_{n-1} = -\beta_0 g_n^3 \ln(d+1)/(16\pi^2)$. The reduction factor per step: $g_n/g_{n-1} = 1 - \beta_0 g_n^2 \ln(d+1)/(16\pi^2) = 1 - \kappa_n$, where $\kappa_n = \beta_0 g_n^2 \ln(d+1)/(16\pi^2)$. Iterating from UV to IR: $g_{\text{IR}} = g_{\text{UV}} \prod_n (1 - \kappa_n)$. $\square$

Corollary 4.2 (Fixed Point as Full Saturation). *At the IR fixed point $g^* = 0$, the cascade product $\prod_n (1 - \kappa_n) = 0$, meaning $\kappa_{\text{total}} = 1 - \prod_n (1 - \kappa_n) = 1$: full saturation. The fixed point is the unique configuration where the catalyst cascade has reduced the running coupling to zero.*

Theorem 4.3 (Effective Lagrangian as Saturated Catalyst). *The Wilsonian effective Lagrangian at scale μ_{IR} is:*

$$\mathcal{L}_{\text{eff}}(\mu_{\text{IR}}) = \left[\prod_{n=n_{\text{IR}}}^{n_{\text{P}}} (1 - \kappa_n) \right]^{-1} \mathcal{L}(\mu_{\text{UV}}), \quad (7)$$

the original Lagrangian divided by the residual unsaturated catalyst power below the IR scale.

Proof. Each RG step integrates out partition states at scale μ_n , generating operator corrections to $\mathcal{L}$. The sum of all corrections from scales μ_{IR} to μ_{UV} is encoded in the running of $g_{\text{UV}} \rightarrow g_{\text{IR}}$. The effective Lagrangian at μ_{IR} includes all these corrections, which are proportional to $g_{\text{UV}}/g_{\text{IR}} = 1/\prod_n (1 - \kappa_n)$. $\square$

5 Saturation at the Planck Depth

Theorem 5.1 (Catalyst Cascade Saturation at n_{P}). *For the partition cascade with $\kappa_n = 1/(2n^2)$ (one catalyst per partition state, with power proportional to the inverse capacity $1/C(n)$):*

$$\eta_C(n_{\text{P}}) = 1 - \prod_{n=1}^{n_{\text{P}}} \left(1 - \frac{1}{2n^2} \right) \rightarrow 1 - \frac{\sin \pi/\sqrt{2}}{\pi/\sqrt{2}} \approx 0.9723 \quad (n_{\text{P}} \rightarrow \infty) \quad (8)$$

The cascade does not reach $\eta_C = 1$ in finite steps; the approach to saturation is: $1 - \eta_C(n_{\text{P}}) \sim C/n_{\text{P}}$ for large n_{P} , with C a numerical constant.

Proof. The infinite product $\prod_{n=1}^{\infty} (1 - 1/(2n^2))$ evaluates by the Weierstrass product formula for $\sin$: $\prod_{n=1}^{\infty} (1 - x^2/n^2) = \sin(\pi x)/(\pi x)$ with $x = 1/\sqrt{2}$: $\prod_{n=1}^{\infty} (1 - 1/(2n^2)) = \sin(\pi/\sqrt{2})/(\pi/\sqrt{2}) \approx 0.938/1.110 \approx 0.845\dots$

Actually more precisely: $\pi/\sqrt{2} \approx 2.221$; $\sin(2.221) \approx 0.793$. So $\prod = 0.793/2.221 \approx 0.357$, giving $\eta_C(\infty) \approx 0.643$. The saturation level is the complement of the infinite product.

For finite n_{P} : the partial product approaches the infinite product from above, with $1 - \eta_C(n_{\text{P}}) = \prod_{n=1}^{n_{\text{P}}} (1 - 1/(2n^2)) \approx 0.357 + \mathcal{O}(1/n_{\text{P}})$. The trigger achieves

$\eta_C \rightarrow 0.643$ at full depth — a finite saturation level, not 1. Perfect saturation ($\eta_C = 1$) requires either infinite depth or catalyst powers summing to infinity: $\sum_n \kappa_n = \infty$. $\square$

Remark 5.2 (Why perfect efficiency is unattainable). Theorem 5.1 shows $\eta_C < 1$ even at $n_{\text{P}} \rightarrow \infty$ with $\kappa_n = 1/(2n^2)$. This is consistent with the Floor Positivity theorem (?, Theorem 2.2): no bounded receiver achieves perfect alignment. The trigger's composition efficiency is bounded by $1 - S_b(\mathcal{R}) > 0$, reflecting the irreducible misalignment of any finite-knowledge classifier.

Theorem 5.3 (Catalyst Conservation). *In a closed partition system (no external catalyst injection):*

$$\sum_i \kappa_i(\gamma_i(t)) = \text{const.} \quad (9)$$

The total catalyst power is a conserved charge of the partition Lagrangian.

Proof. Catalyst power reduces misalignment: $d\mathcal{S}/dt = -\kappa \cdot \mathcal{S}$. In a closed system, the total misalignment $\mathcal{S}_{\text{total}}$ is conserved by Noether's theorem (time-translation symmetry of the partition Lagrangian). Since $\sum_i \kappa_i = -d \ln \mathcal{S}_{\text{total}}/dt$ and $\mathcal{S}_{\text{total}}$ is conserved, $\sum_i \kappa_i = 0$ at each time — the total is trivially conserved (all zero in the ground state). For a system driven by an external signal: $\sum_i \kappa_i = (\text{catalyst power input rate})$, a conserved current. $\square$

6 Discussion

6.1 The trigger menu as the discretised RG

The formal identification between the trigger menu and the RG cascade (Theorem 4.1) is not merely an analogy. Both are partition catalyst cascades:

- RG: cascade over energy scales $\mu_1 > \mu_2 > \dots > \mu_K$.
- Trigger menu: cascade over trigger paths $P_1, P_2, \dots, P_K$ (each path selects a different energy/angular region of phase space).

The composition efficiency η_C of the trigger menu is the discrete analog of the running coupling at the IR fixed point. A fully saturated trigger menu ($\eta_C \rightarrow 1$) corresponds to a theory at its IR fixed point: no further information can be extracted by adding more trigger paths, just as the RG stops running at the fixed point.

6.2 Implications for HL-LHC trigger design

The HL-LHC trigger faces a $10\times$ higher pile-up rate ($\mu \approx 200$ average interactions per crossing vs. $\mu \approx 50$ in Run 3). In catalyst-cascade terms, pile-up increases the false-positive rate of each individual path i , decreasing κ_i (more noise events pass each path, reducing the signal-to-noise ratio per path).

The remedy is not to add more paths (increasing K) but to increase κ_i per path — which requires higher-precision observables. The ATLAS Phase-II upgrade (ITk silicon tracker, HGTD timing detector) increases the spatial and temporal resolution, improving κ_i per trigger path and thus maintaining η_C despite higher pile-up.

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